

AN OVERVIEW OF MICROBIALLY INDUCED CALCITE PRECIPITATION (MICP) TECHNOLOGY

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Abstract

Microbially Induced Calcite Precipitation (MICP) is an innovative biotechnological process that utilizes specific bacteria to precipitate calcium carbonate (CaCO_3), commonly in the form of calcite. This naturally occurring phenomenon has been harnessed in the field of civil engineering to address limitations inherent in traditional cementitious materials. MICP operates by manipulating microbial metabolic processes, where ureolytic bacteria hydrolyze urea to produce carbonate ions, which subsequently react with calcium ions to form calcium carbonate. This calcite precipitation strengthens materials by filling voids, improving mechanical properties, and enhancing durability. MICP has demonstrated its potential in various applications, including soil stabilization, self-healing concrete, and bioremediation of heavy metals. In civil engineering, it strengthens soils by binding particles together and enhances the durability of concrete by enabling autonomous crack healing. Furthermore, MICP offers a sustainable method for environmental remediation, where the biomineralization process immobilizes toxic heavy metals and aids in carbon sequestration by storing CO_2 in mineral forms. This emerging technology represents a significant advancement in modern construction materials, offering eco-friendly alternatives to conventional methods. By integrating lessons from natural calcification processes, MICP has gained increasing attention in recent years. This paper systematically reviews the fundamental principles of MICP, its mechanisms, its engineering properties, and its current applications in civil engineering, showcasing its transformative potential for sustainable infrastructure development.

Keywords: *MICP, Self-healing concrete, Bio-remediation, Soil Improvement*

Introduction

In the face of escalating global environmental crises, sustainable development has emerged as a primary concern for industries worldwide, including civil engineering. Among the key drivers of this shift is the global commitment to achieving the "double carbon" goal, which refers to the reduction of carbon dioxide emissions and the promotion of carbon neutrality. In response to these, there has been an increasing demand for environmentally friendly, energy-saving, and natural materials that can offer long-term sustainability.

Historically, civil engineering has relied on traditional materials like cement and chemical additives for soil reinforcement and concrete repair. Cement, for example, remains a widely used grouting material due to its effectiveness in strengthening foundations. However, cement production is energy-intensive, requiring the burning of fossil fuels, which contributes to high levels of carbon emissions. Moreover, its application in the field often results in environmental degradation. The chemical methods used to reinforce soil or remediate cracks in concrete, such as epoxy resins, sodium silicate, and polyurethane, also introduce toxic byproducts and pollution into ecosystems. These traditional methods not only contribute to environmental harm but also pose practical challenges, including limited long-term durability and susceptibility to secondary cracking in concrete structures. In addition to construction materials, environmental challenges also include the remediation of heavy metal contamination in soils, which arises from industrial activities.

Against this backdrop, Microbially Induced Calcium Carbonate Precipitation (MICP) technology has garnered increasing attention as a potential solution to both construction and environmental challenges. MICP is a natural process wherein microorganisms induce calcium carbonate (CaCO_3) formation through their metabolic activities, which can be harnessed to improve the structural integrity of soils and repair damaged concrete structures.

The underlying mechanism of MICP involves specific bacteria, such as *Sporosarcina pasteurii*, which use urea hydrolysis to produce carbonate ions that combine with calcium ions in the environment to form calcium carbonate crystals. These crystals effectively fill pores in soils or cracks in concrete, resulting in increased stability and strength.

MICP technology offers numerous advantages over traditional construction and remediation methods. The formation of calcium carbonate is self-replicating and regenerative, allowing for continuous reinforcement over time. MICP not only strengthens materials but also enhances their durability, making it an ideal solution for long-term infrastructure projects. Additionally, MICP has been shown to reduce water permeability and increase the unconfined compressive strength of treated soils and concrete, which are critical factors in both geotechnical and structural engineering applications.

MICP has emerged as a promising alternative, offering a more sustainable, eco-friendly approach to construction and environmental remediation. This natural process harnesses the ability of microorganisms to induce calcium carbonate precipitation, which can strengthen soils, stabilize foundations, and repair cracks in concrete structures. It is an attractive option for the remediation of contaminated sites, especially for soils polluted with heavy metals. It aligns with the principles of sustainability, requiring lower energy inputs, reducing environmental impact, and offering adaptable, self-regenerating solutions.

This research report aims to provide an in-depth review of the current advancements in MICP technology. By systematically analyzing these studies, the report will highlight the benefits of MICP in civil engineering applications, such as soil reinforcement, concrete crack repair, and environmental remediation. Additionally, the review will examine the role of different microorganisms and their metabolic pathways.

Literature Review

Javeed et al. 2024 presented a thorough examination of the potential of microbial self-healing technologies in enhancing the durability and sustainability of concrete structures. The paper begins by addressing the common issues faced by concrete structures, such as cracking due to various environmental factors. It highlights that while traditional concrete can last over 50 years, it is prone to cracks that compromise its durability and lead to costly repairs, which can reach up to \$147/m³ annually worldwide. The authors critiqued conventional repair techniques, noting their short-term effectiveness and environmental drawbacks. Traditional methods often fail to address internal damage, leading to a significant percentage of repairs deteriorating within a few years. The paper introduces microbial self-healing as a promising alternative, emphasizing its ability to enhance concrete durability through processes like microbially induced calcium carbonate precipitation (MICP). This biological approach allows for the healing of cracks by utilizing bacteria that can survive in harsh concrete environments. The paper explores various methods for incorporating bacteria into concrete, including direct inclusion and encapsulation as shown in Fig.1. It highlights the challenges associated with high pH levels and the need for carrier technologies to protect bacteria from harsh conditions. A significant portion of the review focuses on the environmental factors affecting bacterial survival, such as pH and temperature. The authors discussed how these factors can limit the effectiveness of bacteria in concrete, necessitating further research into optimal conditions for bacterial activity.

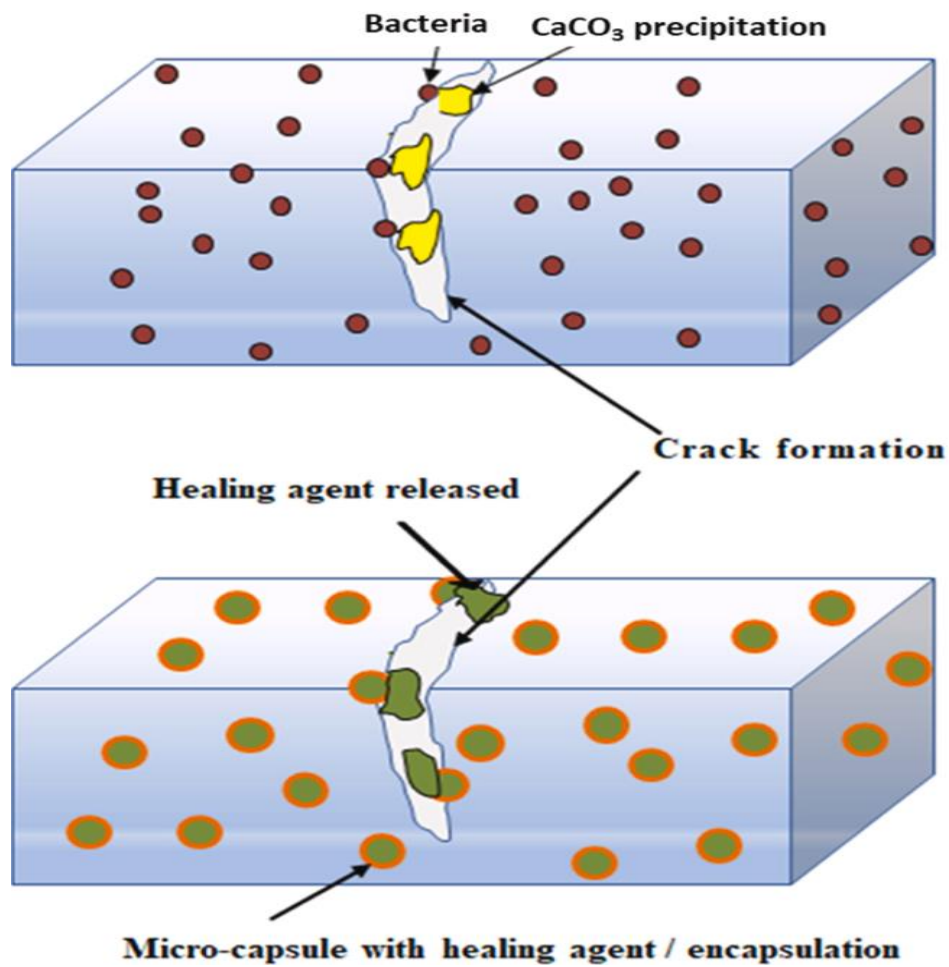


Fig.1. Implementation methods: a) Direct mixing; b) Encapsulation of bacteria (Javeed et al. 2024)

They also examined the impact of different bacterial species on the mechanical properties of concrete, noting significant improvements in strength metrics such as splitting tensile strength and compressive strength.

Jiang et al. 2023 provided a comprehensive overview of the opportunities and challenges associated with Microbial Induced Calcium Carbonate Precipitation (MICP) technology in the field of civil engineering. They highlighted the mechanisms, applications, and future prospects of MICP technology. The paper emphasized the fundamental mechanisms of microbial mineralization, which is crucial for understanding how MICP can be effectively applied in civil engineering. It discussed the metabolic processes of microorganisms that lead to calcium carbonate precipitation, which is essential for enhancing the mechanical properties of soils and rocks. They talked about the four different metabolic mechanisms and pathways: Urease bacteria, Anaerobic denitrifying bacteria, Sulphate-reducing bacteria and Iron reducing bacteria. Table 1 will provide an overview of the primary microbial mechanisms currently recognized for calcium carbonate deposition:

Table 1. Mechanism of MICP (Jiang et al. 2023)

Mechanism of microbial precipitation of calcium carbonate.				
Metabolic pathway	Biochemical reaction	Adaptation to the environment	Representative bacterial type	Application area
Urease bacteria[11]	$CO(NH_2)_2 + 2H_2O \rightarrow H_2CO_3$ $+ 2NH_3$ $H_2CO_3 + 2NH_3 \rightarrow 2NH_4^+ + 2OH^-$ $H_2CO_3 \rightarrow H^+ + HCO_3^-$ $HCO_3^- + H^+ + 2OH^- \rightarrow CO_3^{2-} + 2H_2O$ $Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3 \downarrow$	Aerobic and alkaline environment	<i>Bacillus pasteurii</i> , <i>Sporosarcina pasteurii</i> , <i>Bacillus sphaericus</i> , <i>Bacillus subtilis</i> , <i>Bacillus megaterium</i> , <i>Bacillus diminuta</i> et al.	Soil reinforcement, treatment of liquefied soil, stable slope, repair of concrete cracks, biological bricks, dust prevention and sand fixation, heavy metal pollution control
Anaerobic denitrifying bacteria[7]	$2.6Ca^{2+} + 2CH_3COO^- + 3.2NO_3^-$ $\rightarrow 2.6CaCO_3 \downarrow + 1.6N_2 + 1.4CO_2$	Under anoxic and alkaline conditions	<i>Alcaligenes</i> , <i>Spirillum</i> , <i>Pseudomonas</i> , <i>Achromobacter</i> , <i>Serratia marcescens</i> , <i>Diaphorobacter nitroreducens</i> et al.	Soil reinforcement, repair of concrete cracks, heavy metal pollution control
Sulphate-reducing bacteria[12]	$2CH_2O + OH^- + SO_4^{2-} \rightarrow HS^- + H_2O + 2HCO_3^-$ $HCO_3^- \rightarrow OH^- + CO_2$ $Ca^{2+} + HCO_3^- + OH^- \rightarrow CaCO_3 \downarrow + H_2O$	Anaerobic, alkaline environment	<i>Desulfovibrio</i> , <i>Desulfobacilli</i> , <i>Desulfurmonas</i> , <i>Desulfobulbus</i> , <i>Desulfotomaculum</i> et al.	Soil reinforcement, anti-seepage plugging
Iron-reducing bacteria[13]	$(HCOO)_3Fe + 3OH^- + 3NH_4^+ \rightarrow Fe(OH)_3 \downarrow + 3HCOONH_4$	Aerobic and acidic environment,	<i>Shewanella alga</i> , <i>Shewanella putrefaciens</i> , <i>Stenotrophomonas maltophilia</i> et al.	Soil reinforcement, anti-seepage plugging, heavy metal pollution control

They investigated the engineering properties influenced by MICP, including mechanical strength, permeability, and corrosion resistance. It was found that there exists a positive relationship between the strength of consolidated sand and the content of calcium carbonate (shown in Fig.2.). MICP technique helps to reduce permeability, offers erosion resistance also. The review systematically explores the current applications of MICP in various civil engineering domains. Notable applications include:

Microbial Self-Healing Concrete: MICP technology is utilized to develop concrete that can self-repair cracks, enhancing the longevity of structures.

Soil and Rock Reinforcement: MICP has been successfully applied to consolidate sand columns and improve the integrity of soil.

Cultural Heritage Restoration: The technology is also employed in the restoration of cultural relics, where indigenous bacteria can reinforce stone artifacts without altering their aesthetic appearance.

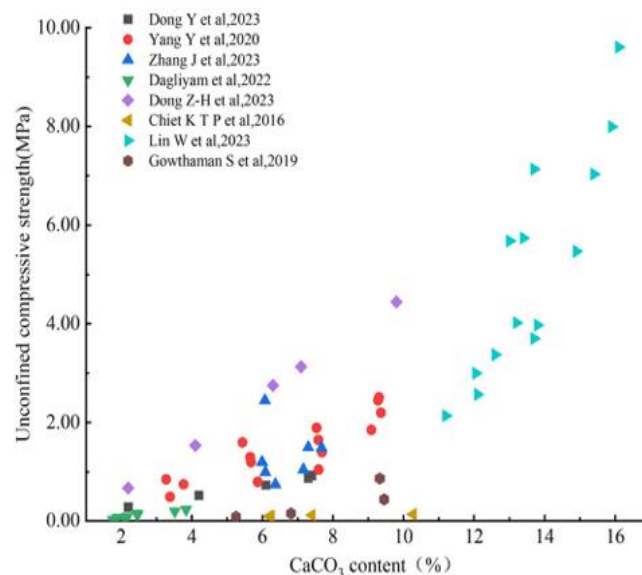


Fig.2. Relationship between unconfined compressive strength and calcium carbonate content of MICP-treated sand. (after Jiang et al. 2023)

Kumar et al. (2023) investigated and discussed the application of MICP in heavy metal removal, mechanism of MICP, and the critical operational factors responsible for efficient microbial activity. The carbonates are produced via three major metabolic pathways of microorganisms: urea hydrolysis, redox reactions, and photosynthesis as shown in Fig.3.

They also discussed about the factors affecting the MICP, i.e. strain, functional genes and proteins, bio stimulation and bio augmentation , and external causes like pH, temperature, concentration of type of heavy metal ions, etc.

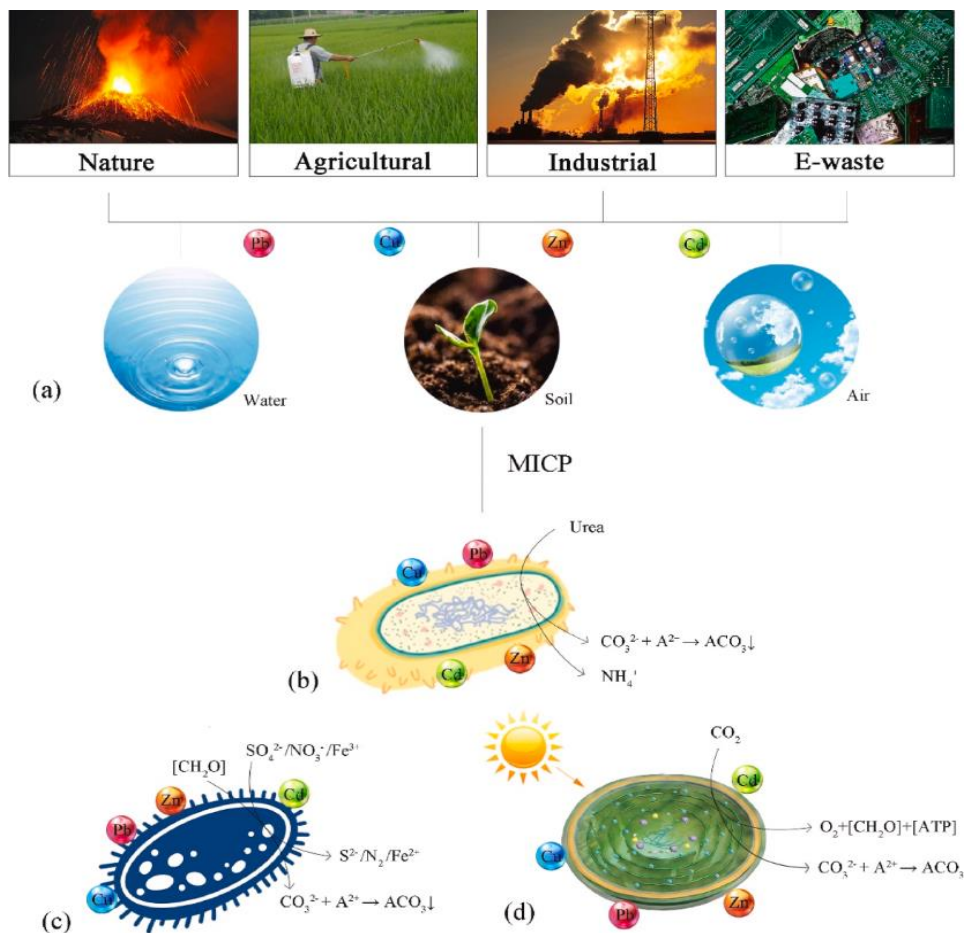


Fig. 3. Sources and mechanisms of carbonate precipitation induced by the microorganism. (Kumar et al. 2023)

Mujah et al. 2016 focused on the history and state-of-the-art of soil improvement using bio cementation through microbially induced calcite precipitation (MICP). The paper specifically discusses the factors affecting the effectiveness of MICP in soil improvement and compares the projected cost incurred by MICP compared to other conventional techniques for field applications. The authors also extensively discussed about the soil treatment process by MICP. The paper stated that the introduction of bacteria to soil can be achieved generally through two main methods including:

(1) injection method and (2) premixing method.

Their research also found that though non uniformity of calcite distribution was found in the soil treated by injection method, still it is the most commonly preferred MICP treatment method as it lies in the injection conditions (i.e. flow, pressure and hydraulic gradient) that can be easily controlled during testing, leaving a room for treatment adjustment which allows injection from both vertical and horizontal directions. Also, this method allows fully saturated, partially saturated and unsaturated conditions of soils to be prepared since the flow of the reactants is controllable. They also mentioned about the different geotechnical properties of bio cemented soils, their advantages, and limitations.

Summary

- MICP is a biological process where bacteria precipitate calcium carbonate (CaCO_3) in concrete. It involves bacteria, typically alkaliphilic or spore-forming strains like *Sporosarcina pasteurii*, which convert urea into calcium carbonate through enzymatic activity, healing cracks in concrete.
- Bacteria can remain dormant and viable in concrete for decades. When cracks form, and water seeps in, the bacteria get activated and produce calcite that seals cracks.
- Applications: Concrete Self-Healing, Bio remediation of heavy metal contaminated soil and water, Soil improvement. It improves Mechanical Strength, Enhances compressive strength (up to 17.8%), flexural strength (29.1%), and tensile strength (35.2%) in concrete.
- Limitations of MICP: Environmental Sensitivity, Challenges in Uniform Distribution: Carrier Technologies Required: Encapsulation techniques, Soil compatibility.

Future Research Areas

Studies need to be conducted on optimizing the microbial strains and enhancing their performance in different environmental conditions. Also currently no study has yet attempted to determine the optimum MICP process in terms of the cost and factors involved, for potential commercial implementation. Further research in terms of upscaling to field applications at the mega level can be also be done. It is also found that this MICP technology is better suited for shallow cracks. Studies may be done on its improvement to repair deep cracks and also on its long-term performance and durability. There is scope of future research on soil compatibility also.

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